# Plan: O6-T — Triangle, the first corner-bearing object (Track A2 probe)

HyMeKo manipulation program

2026-08-12

## 1 Scope and goal

Add **O6-T**, an equilateral triangular prism, as the first *corner-bearing* manipuland in the multi-object curriculum, and characterize it through the existing exact-zero pipeline. This is a **Track A2 (shapes) probe** of the frozen roadmap (the roadmap's `NOW` is Track C0; this is a user-directed shape excursion).

Key fact that makes this cheap: the geometry is **already wired**. `Shape.TRIANGLE`, `ObjectSpec.from_hymeko`, `compose_kwargs`, `footprint_radius`, and the `compose_planar_scene "triangle"` branch (equal-area equilateral prism mesh, density-derived mass/COM/inertia) *all exist*. No new geometry code is written. The object is declared as another HyMeKo scene and added to the curriculum; the existing generation smoke picks it up.

**Why a triangle.** It is the first manipuland with **sharp corners** and a non-box, non-circular cross-section: three flat faces meeting at three vertices, a footprint *circumradius* $R_c = \sqrt{\pi r^2/(\frac{3\sqrt{3}}{4})} \approx 31.1\,\text{mm}$ for the equal-area $r = 0.02$ (vs the coin's 20 mm), and orientation-dependent contact that is neither the disk's antipodal roll nor the box's flat clamp. It exercises the straddle-standoff generalization (`CoinStraddleTargets.for_object(footprint)`) on a genuinely new geometry.

## 2 Affected files (full list)

- **NEW** `data/robotics/galambos_env_o6_triangle.hymeko` — the scene (`@dsk` declares `shape "triangle"`, `radius 0.02`, equal-area, mass = O0).
- **EDIT** `hymeko_rl/coin_delivery/object_curriculum.py` — add `ObjectVariant("O6-T", …, "shape-corner")`; update the module docstring.
- **EDIT** `hymeko_rl/tests/test_object_curriculum.py` — (a) **fix a pre-existing failure**: the membership assertion still reads `["O0","O1-L","O2-M","O4-S"]` although `O5-R` was added earlier (test has been red since R12.2); correct it to the full set incl. `O5-R` and `O6-T`; (b) add a triangle intent test (geom type `MESH`, mass parity to O0, footprint > coin).
- **NEW** `reports/2026-08-12-o6-triangle-shape/` — report + smoke JSON.
- **NEW** `docs/plans/2026-08-12-o6-triangle-shape/` — this plan (4 formats).

No harness is duplicated: the existing `r11_7a_u6a_generation_smoke.py` iterates `U6A_CURRICULUM` and therefore covers O6-T automatically (no smoke edit).

## 3 CORE.YAML items touched

**None.** Verified against `CORE.YAML`: core is the Rust crates (`hymeko_core/query/client/daemon/parser`), `docs/spec/*`, `rtl/*`, and pinned deps. All targets are `hymeko_rl/**` (Python), `data/robotics/*.hymeko`, `tests/**`, `reports/**`, `docs/plans/**` — non-core (`on_unknown_path: treat_as_non_core`). No dependency added or changed.

## 4 Interface changes

None. Additive curriculum entry only; no function signatures change. The scene → ObjectSpec → builder path is unchanged (the triangle branch is pre-existing).

## 5 Test strategy (by layer)

- **Unit** (`test_object_spec.py`, extend if needed): `Shape.from_str("triangle")`, `footprint_radius` circumradius > coin (already present); add a compiled-model mass-parity assertion for a triangle spec.
- **Unit** (`test_object_curriculum.py`): corrected membership; O6-T loads from HyMeKo, keeps the stable `"disk"` handle, geom type `MESH`, mass ≈ O0 (0.050265 kg, equal-area equal-thickness), footprint > O0.
- **Integration**: the generation smoke (`r11_7a_u6a_generation_smoke.py`) over the full curriculum incl. O6-T: 0 model/contract failures, static contracts OK, certified-straddle for the triangle, and per-variant certified-capture counts (the characterization row).
- **Performance**: assert peak RSS < 2 GB (budget; U6A precedent ≈ 312 MB) and record wall time. This is a numerical budget, not a print.

Coverage rule: the O6-T path and the corrected membership are each driven by a new/edited test that would fail against the prior tree (the membership test literally does today).

## 6 Performance budget

Peak RSS < 2 GB (hard cap 16 GB, §4). Wall: generation smoke < 5 min on the Mac CPU runner; plan compile (tectonic) < 30 s. No GPU. Determinism: the smoke fixes seeds (`_SEEDS`); no system entropy.

## 7 Risk anticipation (production-scale)

- **Mass parity.** The mesh mass is density-derived from the convex hull. Risk: the equal-area formula and MuJoCo's volume integration disagree. Mitigation: the intent test asserts `body_mass` ≈ 0.050265 kg within $10^{-4}$; if it fails, the object is not a clean shape-only ablation — halt and report, do not proceed to characterization.
- **Straddle standoff.** The triangle's circumradius footprint (≈ 31 mm) exceeds the coin's (20 mm) and the square's diagonal (≈ 25 mm). The straddle target placement keys off `footprint_radius`; the certified-straddle acquisition may fail more often. This is a *measurement*, not a bug — the smoke's certified-capture count records it, and a low count is an honest characterization result (like O1-L / O4-S), not a failure to fix.
- **Corners vs. contract.** Sharp vertices could produce contact-contract anomalies. The smoke's *0 model/contract failures* gate catches this; if any variant trips it, halt.

- **Pre-existing red test.** `test_object_curriculum.py` is already red (O5-R). This plan fixes it as part of the change; the report declares the fix explicitly (§9 of CLAUDE.md).
- **Worst-case input.** Production scale here is the full curriculum (6 variants × `_POSITIONS` × `_SEEDS`) through the smoke; no training, no overnight run. Well within budget.

# 8 Rollback path

Single commit. Revert = drop the scene file, the `ObjectVariant` line, and the test edits; the curriculum returns to its current 5-entry set and the (still pre-existing-red) membership test is untouched. No migrations, no state mutation, no checkpoints written.

# 9 Out of scope

No teacher-bank generation, no retrieval policy, no RL. If the smoke shows a clean generated triangle with certified captures, a bounded teacher characterization (reach / capture / teacher-K6|capture) may follow as a *separate* step to fill the benchmark row — but only after this gate passes. This plan does not commit to a full R11.7C-style multi-seed campaign.